

Indigo Health Waiting Times – Data Dictionary

Each row relates to the experience of a particular patient with an appointment on a particular day. The last 5 columns (in blue below) are model outcomes based on a two-stage LightGBM algorithm for verification and testing for web prediction engine.

DayOfWeek - The day of the appointment (1 = Monday, 5 = Friday).

TimeOfDay - The time of the appointment. The working day is split up into quarter hour blocks (1 is the first block at 8:00 am, 32 is the last block at 4:45 pm). There is a lunch break from 12 until 1

PatientAge - in years

PatientGender - 0 is Male, 1 is Female

NumOfDoctorsOnDuty - The number of doctors available to see patients on that day.

NumReceptionistsOnDuty - The number of receptionists available to take calls and deal with patients in the waiting room on that day.

Temperature - The outside temperature in degrees Celcius on that day at 12 midday.

Rainfall - The amount of rainfall over the course of the day in millimetres.

PctPriorVacantSlots - The percentage of appointments ("slots") that were vacant before a patient's appointment. A higher percentage of vacant slots can act as a buffer, meaning it is easier to recover when appointments are unexpectedly lengthy or there are emergencies that need to be dealt with. Values range from 0 (0%) to 1 (100%).

PriorEmergency - Whether or not an emergency needed to be dealt with during the day before the patient's appointment (0 = No prior emergency, 1 = Prior emergency)

AvgDoctorExperience - The average experience (in years) of the doctors on duty on that day.

PctSwitch - The percentage of patients on that day who are willing to switch to a different doctor. This can reduce delays if one doctor is running behind schedule whilst another one has free time. Ranges from 0 (0%) to 1 (100%).

AvgAgePrior - The average age of patients seen prior to the appointment (in years).

AvgGenderPrior - The average gender of patients seen prior to the appointment (using the encoding 0 = Male, 1 = Female).

WaitTime - The waiting time experienced by the patient beyond the scheduled start time for their appointment (in minutes).

Source - The model was split into 80% for training, and 20% validation. Values: Train, or Validation

Stage1_Prob_Wait - Probability if the patient had to wait, [0,1]

Stage2_Cond_Time - Expected waiting time (in minutes) given that the patient had to wait

Pred_WaitTime - Final expected waiting time, i.e. $\text{Stage1_Prob_Wait} \times \text{Stage2_Cond_Time}$

Abs_Error - Absolute error of the prediction, i.e. $\text{abs}(\text{WaitTime} - \text{Pred_WaitTime})$